

Rise of the Cosine

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1 Introduction

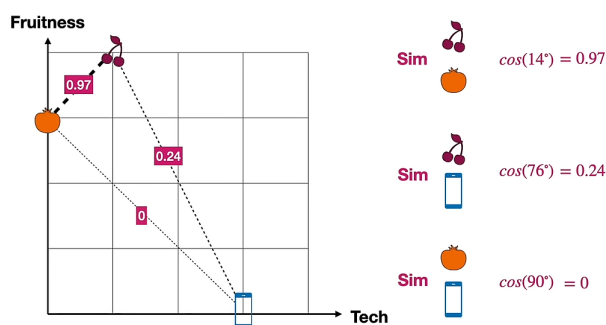
The now classic paper “Attention Is All You Need” laid the conceptual foundation for current implementation of the models such as Chat Generative Pre-trained Transformer (ChatGPT) by employing similarity between words, sentences, and partial paragraphs, in vector calculations called together “attention,” which rely on the cosine¹ function. While reviewing Do-It-Yourself guides for building a model of this type, “generative pre-trained transformer”, the backbone of ChatGPT, I discovered that a very familiar trigonometric function—cosine—was enabling its success. Here, I want to explain the first part of this process (and hope to provide more in later editions) to provide motivation for trigonometry enthusiasts.

2 Steps to Build a Model

A preparatory step in building the model is embedding – representing each word (or root and ending) as a number, and then the whole a sentence as a vector, making it into an array of numbers. In large language models, this provides the link from human speech to the computer.

Although the Euclidean distance could be used as an indicator of relative proximity between two points, using the cosine function has important advantages.

Using words “orange,” “cherry,” and “smartphone” in different sentences leads to determining cosines of angles of respective vectors. Cosine approximates closeness of meaning: 0.97 corresponds to high similarity, while 0 to low.



¹To be fair, sin is also used but the correct term is “cosine similarity.”

In simple terms, softmax replaces cosine with its exponent, making all numbers positive. The maximum is amplified if they are further normalized to sum up to 1.

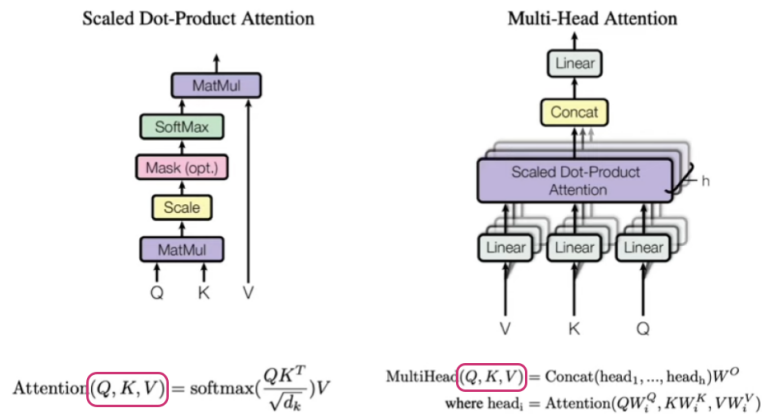
Softmax

$$x \longrightarrow e^x$$

$$\text{Orange} \longrightarrow \frac{e^1 \text{Orange} + e^{0.71} \text{Apple}}{e^1 + e^{0.71}} = 0.57 \text{Orange} + 0.43 \text{Apple}$$

$$\text{! Orange} \longrightarrow \frac{e^1 \text{Orange} + e^{-1} \text{Motorcycle}}{e^1 + e^{-1}} = 0.88 \text{Orange} + 0.12 \text{Motorcycle}$$

The original article by Google researchers uses a more sophisticated formula with a similar meaning:



First, the arrays of numbers built off sentences (embeddings) can be also conceptually likened to vectors in an n -dimensional space. n is the number of words coming together with the one that interests us. Thus, a vector measure is cohesive. The mathematical techniques of vector transformations in linear algebra are well understood, allowing for further work with data in such form. See illustration in Appendix for stretching the angle, and therefore lowering its cosine, between Apple the phone and apple a fruit, with vertex at the origin, by reshaping a square into a narrow and wide parallelogram using linear transformations.

The cosine function has an important advantage as it can be measured for vectors of different dimensions. Other measures such as Euclidean distance, it turned out, tended to increase with the number of dimensions, complicating the task of comparing words in paragraphs and texts of different lengths.

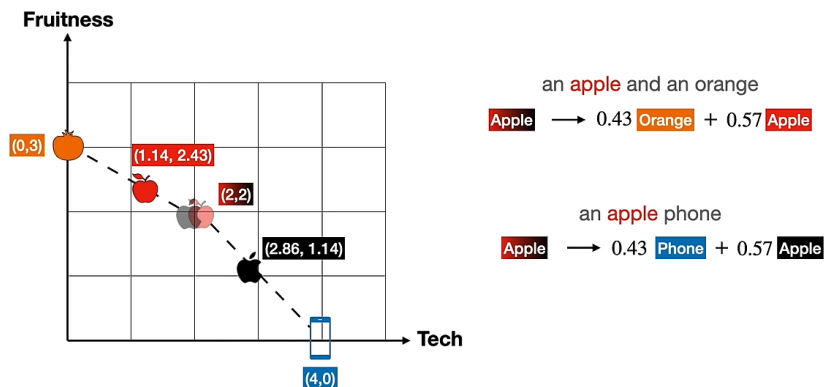
Linked to this property is the fact that cosine's range is $[-1; 1]$, which offers uniformly comparable bounds which make use of words in sentences, paragraphs, and texts of different lengths

comparable. The Euclidean distance is unbounded towards infinity in dimensions and could cause a risk, even for modern computers, of overflow (root calculations are also relatively more “costly” in terms of computing time). This fact can be seen in calculating a distance from the origin to point $(1, 1, 1, 1, \dots, 1)$ in n dimensions: $\sqrt{(1-0)^2 + \dots + (1-0)^2} = \sqrt{n}$, and for $n = 10,000$, corresponding to a smaller dictionary than the one used in ChatGPT, this distance is 100 making this point far away from a “unit circle.”

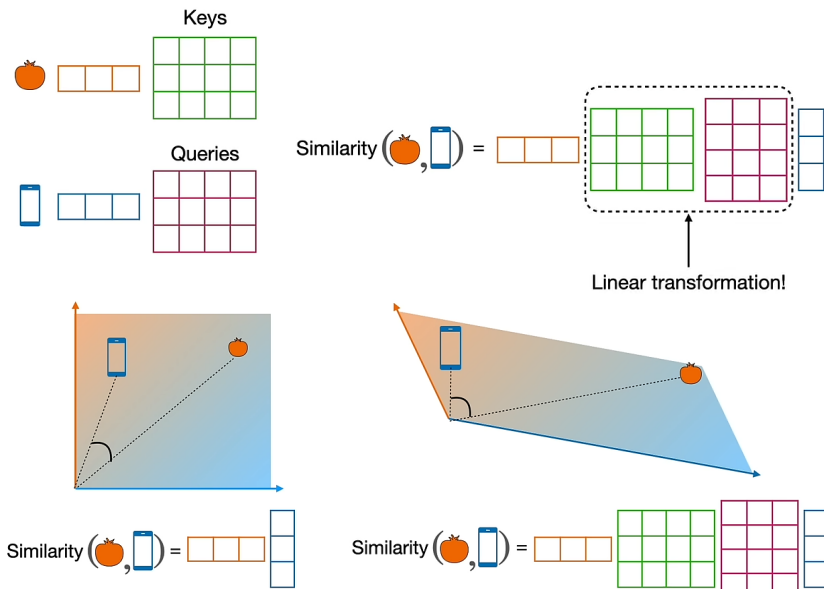
3 Final Adjustment

In the real world, the cosines are replaced with their exponents, as the exponent of a cosine will be a positive number. All of such values lie between $\frac{1}{e}$ (e to power -1) and e (e to power 1). The maximum of several cosines is amplified if exponents are further normalized to sum up to 1, completing the analysis of similarity and finding the likeliest match.

Appendix Illustration: cosine similarity calculation with further transformation



Keys and Queries Matrices



References

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